BradleyGastaldiTerilla2023 — formula report

211 inline formulas, 8 display equations, 1 tables, 13 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
BradleyGastaldiTerilla2023_EQ0008 (4)	007	■ 0.841	`\begin{array}{r} X=\{-, /, \odot, 1,2,3,4,5,6,7,8,9,=, a, b, c, d, e, f, \\ g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z, e \} . \end{array}`	$\begin{array}{r} X=\{-, /, \odot, 1,2,3,4,5,6,7,8,9,=, a, b, c, d, e, f, \\ g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z, e \} . \end{array}$	$X = \{-, /, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, =, \mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \mathtt{e}, \mathtt{f},$ $\mathtt{g}, \mathtt{h}, \mathtt{i}, \mathtt{j}, \mathtt{k}, \mathtt{l}, \mathtt{m}, \mathtt{n}, \mathtt{o}, \mathtt{p}, \mathtt{q}, \mathtt{r}, \mathtt{s}, \mathtt{t}, \mathtt{u}, \mathtt{v}, \mathtt{w}, \mathtt{x}, \mathtt{y}, \mathtt{z}, \acute{\mathtt{e}}\}.$
BradleyGastaldiTerilla2023_EQ0004	004	■ 0.992	`aardvark □ (0.632, 0.370, -0.620, ... , -0.475).`	$aardvark \longmapsto (0.632, 0.370, -0.620, ..., -0.475).$	$\mathtt{aardvark} \mapsto (0.632, 0.370, -0.620, \ldots, -0.475).$
BradleyGastaldiTerilla2023_EQ0001 (1)	003	■ 0.999	`\begin{aligned} & f: \mathbb{R}^{n} \xrightarrow{f_{1}} \mathbb{R}^{n_{1}} \xrightarrow{f_{2}} \mathbb{R}^{n_{2}} \xrightarrow{f_{3}} \cdots \\ & \cdots \xrightarrow{f_{K}} \mathbb{R}^{n_{K}} \xrightarrow{g} \mathbb{R}^{m} \\ & f_{i}(\mathbf{x})=a \left(\mathbf{M}_{i} \mathbf{x}+b_{i}\right), \end{aligned}`	$\begin{aligned} & f: \mathbb{R}^{n} \xrightarrow{f_{1}} \mathbb{R}^{n_{1}} \xrightarrow{f_{2}} \mathbb{R}^{n_{2}} \xrightarrow{f_{3}} \cdots \\ & \cdots \xrightarrow{f_{K}} \mathbb{R}^{n_{K}} \xrightarrow{g} \mathbb{R}^{m} \\ & f_{i}(\mathbf{x})=a \left(\mathbf{M}_{i} \mathbf{x}+b_{i}\right), \end{aligned}$	$f : \mathbb{R}^n \xrightarrow{f_1} \mathbb{R}^{n_1} \xrightarrow{f_2} \mathbb{R}^{n_2} \xrightarrow{f_3} \cdots$ $\cdots \xrightarrow{f_K} \mathbb{R}^{n_K} \xrightarrow{g} \mathbb{R}^m$ $f_i(\mathbf{x}) = a(\mathbf{M}_i\mathbf{x} + b_i),$
BradleyGastaldiTerilla2023_EQ0002	004	■ 0.999	`\begin{gathered} \text { aardvark } \mapsto(1,0,0,0, \ldots, 0,0) \\ \text { aardwolf } \mapsto(0,1,0,0, \ldots, 0,0) \\ \vdots \\ \text { zyzzyva } \mapsto(0,0,0,0, \ldots, 0,1) . \end{gathered}`	$\begin{gathered} \text { aardvark } \mapsto(1,0,0,0, \ldots, 0,0) \\ \text { aardwolf } \mapsto(0,1,0,0, \ldots, 0,0) \\ \vdots \\ \text { zyzzyva } \mapsto(0,0,0,0, \ldots, 0,1) . \end{gathered}$	$\mathtt{aardvark} \mapsto (1, 0, 0, 0, \ldots, 0, 0)$ $\mathtt{aardwolf} \mapsto (0, 1, 0, 0, \ldots, 0, 0)$ $\vdots$ $\mathtt{zyzzyva} \mapsto (0, 0, 0, 0, \ldots, 0, 1).$
BradleyGastaldiTerilla2023_EQ0003 (3)	004	■ 1.000	`D \hookrightarrow \mathbb{R}^{D} \xrightarrow{\sigma} \mathbb{R}^{n_{1}},`	$D \hookrightarrow \mathbb{R}^{D} \xrightarrow{\sigma} \mathbb{R}^{n_{1}},$	$D \hookrightarrow \mathbb{R}^D \xrightarrow{\sigma} \mathbb{R}^{n_1},$
BradleyGastaldiTerilla2023_EQ0005	005	■ 1.000	`M^{*} u_{j}=\sigma_{j} v_{j} \text { and } M v_{j}= \sigma_{j} u_{j} .`	$M^{*} u_{j}=\sigma_{j} v_{j} \text { and } M v_{j}= \sigma_{j} u_{j} .$	$M^*u_j = \sigma_j v_j$ and $Mv_j = \sigma_j u_j.$
BradleyGastaldiTerilla2023_EQ0006	006	■ 1.000	`F^{*} c_{i} \cong d_{i} \text { and } F_{*} d_{i} \cong c_{i} .`	$F^{*} c_{i} \cong d_{i} \text { and } F_{*} d_{i} \cong c_{i} .$	$F^*c_i \cong d_i$ and $F_*d_i \cong c_i.$
BradleyGastaldiTerilla2023_EQ0007	006	■ 1.000	`R^{*}\left(A_{i}\right)=B_{i} \text { and } R_{*} \left(B_{i}\right)=A_{i}`	$R^{*}\left(A_{i}\right)=B_{i} \text { and } R_{*} \left(B_{i}\right)=A_{i}$	$R^*(A_i) = B_i$ and $R_*(B_i) = A_i$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Inline formulas (first occurrence)

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th></tr>
<tr><td>BradleyGastaldiTerilla2023_F00001</td><td>002</td><td>—</td><td>X</td><td>X</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00003</td><td>002</td><td>—</td><td>\operatorname{Fun}(X)</td><td>$\operatorname{Fun}(X)$</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00005</td><td>002</td><td>—</td><td>k</td><td>k</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00006</td><td>002</td><td>—</td><td>X \rightarrow k^{X}</td><td>$X \rightarrow k^{X}$</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00007</td><td>002</td><td>—</td><td>x \in X</td><td>$x \in X$</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00009</td><td>002</td><td>—</td><td>y \mapsto \delta_{x y}</td><td>$y \mapsto \delta_{x y}$</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00010</td><td>002</td><td>—</td><td>\langle f \mid g\rangle=\sum_{x \in X} f(x) g(x)</td><td>$\langle f \mid g\rangle=\sum_{x \in X} f(x) g(x)$</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00011</td><td>002</td><td>—</td><td>|x\rangle</td><td>$|x\rangle$</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00012</td><td>002</td><td>—</td><td>|x\rangle \in k^{X}</td><td>$|x\rangle \in k^{X}$</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00015</td><td>002</td><td>—</td><td>\{|0\rangle,|1\rangle\}</td><td>$\{|0\rangle,|1\rangle\}$</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00019</td><td>002</td><td>—</td><td>y</td><td>y</td></tr>
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<tr><td>BradleyGastaldiTerilla2023_F00024</td><td>002</td><td>—</td><td>v \wedge w</td><td>$v \wedge w$</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00025</td><td>002</td><td>—</td><td>v</td><td>v</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00026</td><td>002</td><td>—</td><td>w</td><td>w</td></tr>
<tr><td>BradleyGastaldiTerilla2023_F00027</td><td>002</td><td>—</td><td>0 \leftrightarrow 1</td><td>$0 \leftrightarrow 1$</td></tr>
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BradleyGastaldiTerilla2023_F00028	002	—	`{ }^{\text {CP }}`	${ }^{\text {CP }}$
BradleyGastaldiTerilla2023_F00029	002	—	`F`	F
BradleyGastaldiTerilla2023_F00030	002	—	`{ }^{\text {cop }}`	${ }^{\text {cop }}$
BradleyGastaldiTerilla2023_F00031	002	—	`{ }^{\text {COP }}`	${ }^{\text {COP }}$
BradleyGastaldiTerilla2023_F00032	002	—	`{ }^{\mathrm{c} \text { op }}`	${ }^{\mathrm{c} \text { op }}$
BradleyGastaldiTerilla2023_F00033	002	—	`h^{x}`	h^{x}
BradleyGastaldiTerilla2023_F00034	002	—	`h^{x}:=\mathrm{C}(-, x)`	$h^{x}:=\mathrm{C}(-, x)$
BradleyGastaldiTerilla2023_F00035	002	—	`\mathrm{C}(y, x)`	$\mathrm{C}(y, x)$
BradleyGastaldiTerilla2023_F00036	002	—	`x \mapsto h^{x}`	$x \mapsto h^{x}$
BradleyGastaldiTerilla2023_F00037	002	—	`{ }^{\text {Cop }}`	${ }^{\text {Cop }}$
BradleyGastaldiTerilla2023_F00038	002	—	`\mathrm{X}=\mathrm{X}^{\mathrm{op}}`	$\mathrm{X}=\mathrm{X}^{\mathrm{op}}$
BradleyGastaldiTerilla2023_F00039	002	—	`F(x)`	$F(x)$
BradleyGastaldiTerilla2023_F00040	002	—	`\emptyset`	$\emptyset$
BradleyGastaldiTerilla2023_F00041	002	—	`x \amalg y`	$x \amalg y$
BradleyGastaldiTerilla2023_F00042	002	—	`h^{x} \coprod h^{y}`	$h^{x} \coprod h^{y}$
BradleyGastaldiTerilla2023_F00043	002	—	`h^{y}`	h^{y}
BradleyGastaldiTerilla2023_F00044	002	—	`*`	$*$
BradleyGastaldiTerilla2023_F00045	002	—	`h^{x} \coprod h^{x}`	$h^{x} \coprod h^{x}$
BradleyGastaldiTerilla2023_F00046	002	—	`* \sqcup *`	$* \sqcup *$
BradleyGastaldiTerilla2023_F00047	002	—	`\mathrm{X}^{\mathrm{op}} \rightarrow`	$\mathrm{X}^{\mathrm{op}} \rightarrow$
BradleyGastaldiTerilla2023_F00048	002	—	`{ }^{\text {Xop }}`	${ }^{\text {Xop }}$
BradleyGastaldiTerilla2023_F00049	002	—	`0 \rightarrow 1`	$0 \rightarrow 1$
BradleyGastaldiTerilla2023_F00050	002	—	`2^{\text {CP }}`	$2^{\text {CP }}$
BradleyGastaldiTerilla2023_F00051	003	—	`2^{\mathrm{X}}`	2^{X}
BradleyGastaldiTerilla2023_F00052	003	—	`f`	f
BradleyGastaldiTerilla2023_F00053	003	—	`g`	g
BradleyGastaldiTerilla2023_F00054	003	—	`f \vee g`	$f \vee g$
BradleyGastaldiTerilla2023_F00055	003	—	`f \wedge g`	$f \wedge g$

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BradleyGastaldiTerilla2023_FO0056	003	—	`F_{2}=\{0,1\}`	$F_{2}=\{0,1\}$
BradleyGastaldiTerilla2023_FO0057	003	—	`2=\{0,1\}`	$2=\{0,1\}$
BradleyGastaldiTerilla2023_FO0058	003	—	`i n`	$i n$
BradleyGastaldiTerilla2023_FO0059	003	—	`G`	G
BradleyGastaldiTerilla2023_FO0060	003	—	`\mathbb{C}^{G}`	$\mathbb{C}^{G}$
BradleyGastaldiTerilla2023_FO0061	003	—	`\mathbb{C}^{G} \rightarrow \mathbb{C}^{G}`	$\mathbb{C}^{G} \rightarrow \mathbb{C}^{G}$
BradleyGastaldiTerilla2023_FO0062	003	—	`f: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}`	$f: \mathbb{R}^{n} \rightarrow \mathbb{R}^{m}$
BradleyGastaldiTerilla2023_FO0063	003	—	`\mathbf{M}_{i}`	$\mathbf{M}_{i}$
BradleyGastaldiTerilla2023_FO0064	003	—	`n_{i} \times n_{i-1}`	$n_{i} \times n_{i-1}$
BradleyGastaldiTerilla2023_FO0065	003	—	`b_{i} \in \mathbb{R}^{n_{i}}`	$b_{i} \in \mathbb{R}^{n_{i}}$
BradleyGastaldiTerilla2023_FO0066	003	—	`a`	a
BradleyGastaldiTerilla2023_FO0067	003	—	`M_{i}`	M_{i}
BradleyGastaldiTerilla2023_FO0068	003	—	`b_{i}`	b_{i}
BradleyGastaldiTerilla2023_FO0069	003	—	`D`	D
BradleyGastaldiTerilla2023_FO0070	004	—	`D \rightarrow \mathbb{R}^{D}`	$D \rightarrow \mathbb{R}^{D}$
BradleyGastaldiTerilla2023_FO0071	004	—	`\mathbb{R}^{D}`	$\mathbb{R}^{D}$
BradleyGastaldiTerilla2023_FO0072	004	—	`f_{1}`	f_{1}
BradleyGastaldiTerilla2023_FO0073	004	—	`\sigma`	σ
BradleyGastaldiTerilla2023_FO0074	004	—	`\mathbb{R}^{n_{1}}`	$\mathbb{R}^{n_{1}}$
BradleyGastaldiTerilla2023_FO0075	004	—	`\sigma: \mathbb{R}^{D} \rightarrow \mathbb{R}^{n_{1}}`	$\sigma: \mathbb{R}^{D} \rightarrow \mathbb{R}^{n_{1}}$
BradleyGastaldiTerilla2023_FO0076	004	—	`|D| \times|D|`	$\|D\| \times\|D\|$
BradleyGastaldiTerilla2023_FO0077	004	—	`M`	M
BradleyGastaldiTerilla2023_FO0078	004	—	`i`	i
BradleyGastaldiTerilla2023_FO0079	004	—	`j`	j
BradleyGastaldiTerilla2023_FO0080	004	—	`w_{i}`	w_{i}
BradleyGastaldiTerilla2023_FO0081	004	—	`w_{j}`	w_{j}
BradleyGastaldiTerilla2023_FO0082	005	—	`\sigma^{\prime}`	$\sigma^{\prime}$
BradleyGastaldiTerilla2023_FO0083	005	—	`|D| \times d`	$\|D\| \times d$
BradleyGastaldiTerilla2023_FO0084	005	—	`d \times|D|`	$d \times\|D\|$

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BradleyGastaldiTerilla2023_F00085	005	—	`d \ll|D|`	$d \ll|D|$
BradleyGastaldiTerilla2023_F00086	005	—	`\left|M-\sigma^{\prime} \sigma\right|`	$\left\|M-\sigma^{\prime} \sigma\right\|$
BradleyGastaldiTerilla2023_F00087	005	—	`Y`	Y
BradleyGastaldiTerilla2023_F00088	005	—	`X-Y`	$X-Y$
BradleyGastaldiTerilla2023_F00089	005	—	`m: X \times Y \rightarrow k`	$m: X \times Y \rightarrow k$
BradleyGastaldiTerilla2023_F00090	005	—	`m`	m
BradleyGastaldiTerilla2023_F00091	005	—	`X \rightarrow k^{Y}`	$X \rightarrow k^{Y}$
BradleyGastaldiTerilla2023_F00092	005	—	`Y \rightarrow k^{X}`	$Y \rightarrow k^{X}$
BradleyGastaldiTerilla2023_F00093	005	—	`x \mapsto m(x,-)`	$x \mapsto m(x,-)$
BradleyGastaldiTerilla2023_F00094	005	—	`y \mapsto m(-, y)`	$y \mapsto m(-, y)$
BradleyGastaldiTerilla2023_F00095	005	—	`|X|`	$|X|$
BradleyGastaldiTerilla2023_F00096	005	—	`|Y|`	$|Y|$
BradleyGastaldiTerilla2023_F00097	005	—	`m(x, y)`	$m(x, y)$
BradleyGastaldiTerilla2023_F00098	005	—	`m(x,-) \in k^{Y}`	$m(x,-) \in k^{Y}$
BradleyGastaldiTerilla2023_F00099	005	—	`m(-, y) \in k^{X}`	$m(-, y) \in k^{X}$
BradleyGastaldiTerilla2023_F00100	005	—	`M^{*}: k^{X} \rightarrow k^{Y}`	$M^{*}: k^{X} \rightarrow k^{Y}$
BradleyGastaldiTerilla2023_F00101	005	—	`M: k^{Y} \rightarrow k^{X}`	$M: k^{Y} \rightarrow k^{X}$
BradleyGastaldiTerilla2023_F00102	005	—	`M^{*}`	M^{*}
BradleyGastaldiTerilla2023_F00103	005	—	`M M^{*}: k^{X} \rightarrow k^{X}`	$M M^{*}: k^{X} \rightarrow k^{X}$
BradleyGastaldiTerilla2023_F00104	005	—	`M^{*} M: k^{Y} \rightarrow k^{Y}`	$M^{*} M: k^{Y} \rightarrow k^{Y}$
BradleyGastaldiTerilla2023_F00105	005	—	`\mathbb{R}`	$\mathbb{R}$
BradleyGastaldiTerilla2023_F00106	005	—	`\left\{u_{1}, \ldots, u_{m}\right\}`	$\left\{u_{1}, \ldots, u_{m}\right\}$
BradleyGastaldiTerilla2023_F00107	005	—	`\left\{v_{1}, \ldots, v_{n}\right\}`	$\left\{v_{1}, \ldots, v_{n}\right\}$
BradleyGastaldiTerilla2023_F00108	005	—	`k^{Y}`	k^{Y}
BradleyGastaldiTerilla2023_F00109	005	—	`M M^{*}`	$M M^{*}$
BradleyGastaldiTerilla2023_F00110	005	—	`M^{*} M`	$M^{*} M$
BradleyGastaldiTerilla2023_F00111	005	—	`\left\{\lambda_{1}, \ldots, \lambda_{r}, 0, \ldots, 0\right\}`	$\left\{\lambda_{1}, \ldots, \lambda_{r}, 0, \ldots, 0\right\}$
BradleyGastaldiTerilla2023_F00112	005	—	`r=\min (m, n)`	$r=\min (m, n)$
BradleyGastaldiTerilla2023_F00113	005	—	`M=U \Sigma V^{*}`	$M=U \Sigma V^{*}$

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Identifier	Page	Conf.	LaTeX source	Rendered
BradleyGastaldiTerilla2023_F00114	005	—	`\left\{u_{i}\right\}`	$\left\{u_{i}\right\}$
BradleyGastaldiTerilla2023_F00115	005	—	`U`	U
BradleyGastaldiTerilla2023_F00116	005	—	`\left\{v_{j}\right\}`	$\left\{v_{j}\right\}$
BradleyGastaldiTerilla2023_F00117	005	—	`V^{*}`	V^{*}
BradleyGastaldiTerilla2023_F00118	005	—	`\Sigma`	Σ
BradleyGastaldiTerilla2023_F00119	005	—	`m \times n`	$m \times n$
BradleyGastaldiTerilla2023_F00120	005	—	`\sigma_{i}=\sqrt{\lambda_{i}}`	$\sigma_{i}=\sqrt{\lambda_{i}}$
BradleyGastaldiTerilla2023_F00121	005	—	`V`	V
BradleyGastaldiTerilla2023_F00122	005	—	`U^{*} U=I`	$U^{*} U=I$
BradleyGastaldiTerilla2023_F00123	005	—	`V^{*} V=I`	$V^{*} V=I$
BradleyGastaldiTerilla2023_F00124	005	—	`\sigma_{i}`	σ_{i}
BradleyGastaldiTerilla2023_F00125	005	—	`\left\{u_{1}, \ldots, u_{r}\right\}`	$\left\{u_{1}, \ldots, u_{r}\right\}$
BradleyGastaldiTerilla2023_F00126	005	—	`\left\{v_{1}, \ldots, v_{r}\right\}`	$\left\{v_{1}, \ldots, v_{r}\right\}$
BradleyGastaldiTerilla2023_F00127	005	—	`\left\{\left(u_{1}, v_{1}\right), \ldots,\left(u_{r}, v_{r}\right)\right\}`	$\left\{\left(u_{1}, v_{1}\right), \ldots,\left(u_{r}, v_{r}\right)\right\}$
BradleyGastaldiTerilla2023_F00128	005	—	`k^{X} \times k^{Y}`	$k^{X} \times k^{Y}$
BradleyGastaldiTerilla2023_F00129	005	—	`\left(u_{i}, v_{i}\right) \leq\left(u_{j}, v_{j}\right)`	$\left(u_{i}, v_{i}\right) \leq\left(u_{j}, v_{j}\right)$
BradleyGastaldiTerilla2023_F00130	005	—	`\sigma_{i} \leq \sigma_{j}`	$\sigma_{i} \leq \sigma_{j}$
BradleyGastaldiTerilla2023_F00131	005	—	`M^{\prime}`	$M^{\prime}$
BradleyGastaldiTerilla2023_F00132	005	—	`s`	s
BradleyGastaldiTerilla2023_F00133	005	—	`M^{\prime}=U \Sigma^{\prime} V^{*}`	$M^{\prime}=U \Sigma^{\prime} V^{*}$
BradleyGastaldiTerilla2023_F00134	005	—	`\Sigma^{\prime}`	$\Sigma^{\prime}$
BradleyGastaldiTerilla2023_F00135	005	—	`U, \Sigma^{\prime}`	$U, \Sigma^{\prime}$
BradleyGastaldiTerilla2023_F00136	005	—	`U \Sigma^{\prime} V^{*}`	$U \Sigma^{\prime} V^{*}$
BradleyGastaldiTerilla2023_F00137	005	—	`M^{\prime}=U^{\prime} \Sigma^{\prime \prime} V^{\prime *} \approx M`	$M^{\prime}=U^{\prime} \Sigma^{\prime \prime} V^{\prime *} \approx M$
BradleyGastaldiTerilla2023_F00138	005	—	`U^{\prime}`	$U^{\prime}$
BradleyGastaldiTerilla2023_F00139	005	—	`m \times s`	$m \times s$
BradleyGastaldiTerilla2023_F00140	005	—	`V^{\prime *}`	$V^{\prime *}$
BradleyGastaldiTerilla2023_F00141	005	—	`s \times n`	$s \times n$
BradleyGastaldiTerilla2023_F00142	005	—	`\Sigma^{\prime \prime}`	$\Sigma^{\prime \prime}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
BradleyGastaldiTerilla2023_F00143	005	—	`s \times s`	$s \times s$
BradleyGastaldiTerilla2023_F00144	006	—	`f: \mathrm{C}^{\mathrm{op}} \times \mathrm{D} \rightarrow`	$f: \mathrm{C}^{\mathrm{op}} \times \mathrm{D} \rightarrow$
BradleyGastaldiTerilla2023_F00145	006	—	`\mathrm{C} \rightarrow\left(\text { Set }^{\mathrm{D}}\right)^{\mathrm{op}}`	$\mathrm{C} \rightarrow\left(\text { Set }^{\mathrm{D}}\right)^{\mathrm{op}}$
BradleyGastaldiTerilla2023_F00146	006	—	`\mathrm{D} \rightarrow`	$\mathrm{D} \rightarrow$
BradleyGastaldiTerilla2023_F00147	006	—	`^{\mathrm{C}^{\text {Op }}}`	$^{\mathrm{C}^{\text {Op }}}$
BradleyGastaldiTerilla2023_F00148	006	—	`c \mapsto f(c,-)`	$c \mapsto f(c,-)$
BradleyGastaldiTerilla2023_F00149	006	—	`d \mapsto f(-, d)`	$d \mapsto f(-, d)$
BradleyGastaldiTerilla2023_F00150	006	—	`f(c,-)`	$f(c,-)$
BradleyGastaldiTerilla2023_F00151	006	—	`c`	c
BradleyGastaldiTerilla2023_F00152	006	—	`f(-, d): \mathrm{C}^{\mathrm{op}} \rightarrow`	$f(-, d): \mathrm{C}^{\mathrm{op}} \rightarrow$
BradleyGastaldiTerilla2023_F00153	006	—	`d`	d
BradleyGastaldiTerilla2023_F00154	006	—	`\mathrm{C} \rightarrow\left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}`	$\mathrm{C} \rightarrow\left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$
BradleyGastaldiTerilla2023_F00155	006	—	`{ }^{\mathrm{Cop}}`	${ }^{\mathrm{Cop}}$
BradleyGastaldiTerilla2023_F00156	006	—	`F^{*}`	F^{*}
BradleyGastaldiTerilla2023_F00157	006	—	`{ }^{\text {Cop }} \rightarrow\left(\text { Set }^{\mathrm{D}}\right)^{\text {op }}`	${ }^{\text {Cop }} \rightarrow\left(\text { Set }^{\mathrm{D}}\right)^{\text {op }}$
BradleyGastaldiTerilla2023_F00158	006	—	`F_{*}:\left(\text { Set }^{\mathrm{D}}\right)^{\mathrm{op}} \rightarrow`	$F_{*}:\left(\text { Set }^{\mathrm{D}}\right)^{\mathrm{op}} \rightarrow$
BradleyGastaldiTerilla2023_F00159	006	—	`^{\mathrm{C}^{\mathrm{op}}}`	$^{\mathrm{C}^{\mathrm{op}}}$
BradleyGastaldiTerilla2023_F00160	006	—	`F_{*}`	F_{*}
BradleyGastaldiTerilla2023_F00161	006	—	`{ }^{\mathrm{D}}`	${ }^{\mathrm{D}}$
BradleyGastaldiTerilla2023_F00162	006	—	`F^{*} F_{*}`	$F^{*} F_{*}$
BradleyGastaldiTerilla2023_F00163	006	—	`F_{*} F^{*}`	$F_{*} F^{*}$
BradleyGastaldiTerilla2023_F00164	006	—	`c_{i}, d_{i}`	c_{i}, d_{i}
BradleyGastaldiTerilla2023_F00165	006	—	`\mathrm{C}^{\mathrm{op}} \times \mathrm{D}`	$\mathrm{C}^{\mathrm{op}} \times \mathrm{D}$
BradleyGastaldiTerilla2023_F00166	006	—	`\left\{\left(c_{i}, d_{i}\right)\right\}`	$\left\{\left(c_{i}, d_{i}\right)\right\}$
BradleyGastaldiTerilla2023_F00167	006	—	`\left\{\left(u_{i}, v_{i}\right)\right\}`	$\left\{\left(u_{i}, v_{i}\right)\right\}$
BradleyGastaldiTerilla2023_F00168	006	—	`r`	r
BradleyGastaldiTerilla2023_F00169	006	—	`r: X \times Y \rightarrow\{0,1\}`	$r: X \times Y \rightarrow\{0,1\}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
BradleyGastaldiTerilla2023_F00170	006	—	`X \times Y`	$X \times Y$
BradleyGastaldiTerilla2023_F00171	006	—	`R^{*}: 2^{X} \rightarrow 2^{Y}`	$R^{*}: 2^{X} \rightarrow 2^{Y}$
BradleyGastaldiTerilla2023_F00172	006	—	`R_{*}: 2^{Y} \rightarrow 2^{X}`	$R_{*}: 2^{Y} \rightarrow 2^{X}$
BradleyGastaldiTerilla2023_F00173	006	—	`R^{*}`	R^{*}
BradleyGastaldiTerilla2023_F00174	006	—	`A \subseteq X`	$A \subseteq X$
BradleyGastaldiTerilla2023_F00175	006	—	`R^{*}(A)=\{y \in Y: R(x, y)=1`	$R^{*}(A)=\{y \in Y: R(x, y)=1$
BradleyGastaldiTerilla2023_F00176	006	—	`x \in A\}`	$x \in A\}$
BradleyGastaldiTerilla2023_F00177	006	—	`R_{*}`	R_{*}
BradleyGastaldiTerilla2023_F00178	006	—	`B \subseteq Y`	$B \subseteq Y$
BradleyGastaldiTerilla2023_F00179	006	—	`R_{*}(B)=\{x \in X: R(x, y)=`	$R_{*}(B)=\{x \in X: R(x, y)=$
BradleyGastaldiTerilla2023_F00180	006	—	`y \in B\}`	$y \in B\}$
BradleyGastaldiTerilla2023_F00181	006	—	`R^{*} R_{*}`	$R^{*} R_{*}$
BradleyGastaldiTerilla2023_F00182	006	—	`R_{*} R^{*}`	$R_{*} R^{*}$
BradleyGastaldiTerilla2023_F00183	006	—	`\left(A_{i}, B_{i}\right) \subset X \times Y`	$\left(A_{i}, B_{i}\right) \subset X \times Y$
BradleyGastaldiTerilla2023_F00184	006	—	`\left\{\left(A_{i}, B_{i}\right)\right\}`	$\left\{\left(A_{i}, B_{i}\right)\right\}$
BradleyGastaldiTerilla2023_F00185	006	—	`\left(A_{i}, B_{i}\right) \leq\left(A_{j}, B_{j}\right)`	$\left(A_{i}, B_{i}\right) \leq\left(A_{j}, B_{j}\right)$
BradleyGastaldiTerilla2023_F00186	006	—	`A_{i} \subseteq A_{j}`	$A_{i} \subseteq A_{j}$
BradleyGastaldiTerilla2023_F00187	006	—	`B_{i} \supseteq B_{j}`	$B_{i} \supseteq B_{j}$
BradleyGastaldiTerilla2023_F00188	006	—	`\left(A_{i}, B_{j}\right) \wedge\left(A_{j}, B_{j}\right):=\left(A_{i} \cap A_{j}, R^{*} R_{*} \left(B_{i} \cup B_{j}\right)\right)`	$\left(A_{i}, B_{j}\right) \wedge\left(A_{j}, B_{j}\right):=\left(A_{i} \cap A_{j}, R^{*} R_{*} \left(B_{i} \cup B_{j}\right)\right)$
BradleyGastaldiTerilla2023_F00189	006	—	`\left(A_{i}, B_{j}\right) \vee\left(A_{j}, B_{j}\right):=\left(R_{*} R^{*}\left(A_{i} \cup A_{j} \right), B_{i} \cap B_{j}\right)`	$\left(A_{i}, B_{j}\right) \vee\left(A_{j}, B_{j}\right):=\left(R_{*} R^{*}\left(A_{i} \cup A_{j} \right), B_{i} \cap B_{j}\right)$
BradleyGastaldiTerilla2023_F00190	007	—	`Y=X \times X`	$Y=X \times X$
BradleyGastaldiTerilla2023_F00191	007	—	`m: X \times Y \rightarrow \mathbb{R}`	$m: X \times Y \rightarrow \mathbb{R}$
BradleyGastaldiTerilla2023_F00192	007	—	`\left(y_{l}, y_{r}\right) \in Y`	$\left(y_{l}, y_{r}\right) \in Y$
BradleyGastaldiTerilla2023_F00193	007	—	`m(\mathrm{~h},(\mathrm{t}, \mathrm{e})) \approx`	$m(\mathrm{~h},(\mathrm{t}, \mathrm{e})) \approx$
BradleyGastaldiTerilla2023_F00194	007	—	`m(\mathrm{~h},(\mathrm{p}, \mathrm{o})) \approx 0.0037`	$m(\mathrm{~h},(\mathrm{p}, \mathrm{o})) \approx 0.0037$
BradleyGastaldiTerilla2023_F00195	007	—	`X \rightarrow \mathbb{R}^{Y}`	$X \rightarrow \mathbb{R}^{Y}$
BradleyGastaldiTerilla2023_F00196	007	—	`M^{*}: \mathbb{R}^{X} \rightarrow \mathbb{R}^{Y}`	$M^{*}: \mathbb{R}^{X} \rightarrow \mathbb{R}^{Y}$
BradleyGastaldiTerilla2023_F00197	008	—	`\{(A, B)\}`	$\{(A, B)\}$
BradleyGastaldiTerilla2023_F00198	008	—	`A`	A

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Identifier	Page	Conf.	LaTeX source	Rendered
BradleyGastaldiTerilla2023_FO0199	008	—	`|B| \geq 20`	$|B| \geq 20$
BradleyGastaldiTerilla2023_FO0200	008	—	`A_{i}`	A_{i}
BradleyGastaldiTerilla2023_FO0201	008	—	`B_{i}`	B_{i}
BradleyGastaldiTerilla2023_FO0202	008	—	`\vee`	$\vee$
BradleyGastaldiTerilla2023_FO0203	008	—	`\wedge`	$\wedge$
BradleyGastaldiTerilla2023_FO0204	008	—	`y_{l}, y_{r}`	y_{l}, y_{r}
BradleyGastaldiTerilla2023_FO0205	011	—	`[-\infty, \infty]`	$[-\infty, \infty]$
BradleyGastaldiTerilla2023_FO0206	011	—	`[-\infty, \infty]^{X}`	$[-\infty, \infty]^{X}$
BradleyGastaldiTerilla2023_FO0207	011	—	`2^{X}`	2^{X}
BradleyGastaldiTerilla2023_FO0208	011	—	`\{0 \rightarrow 1\}`	$\{0 \rightarrow 1\}$
BradleyGastaldiTerilla2023_FO0209	011	—	`V_{1} \otimes V_{2} \otimes \cdots \otimes V_{n}`	$V_{1} \otimes V_{2} \otimes \cdots \otimes V_{n}$
BradleyGastaldiTerilla2023_FO0210	011	—	`n-1`	$n-1$
BradleyGastaldiTerilla2023_FO0211	011	—	`\mathrm{C}_{1} \times \mathrm{C}_{2} \times \cdots \times \mathrm{C}_{n}`	$\mathrm{C}_{1} \times \mathrm{C}_{2} \times \cdots \times \mathrm{C}_{n}$

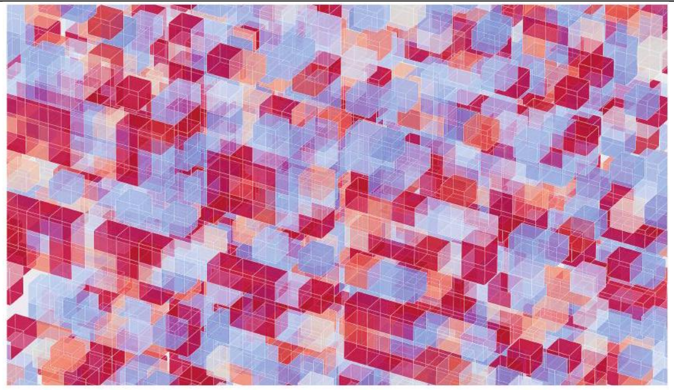
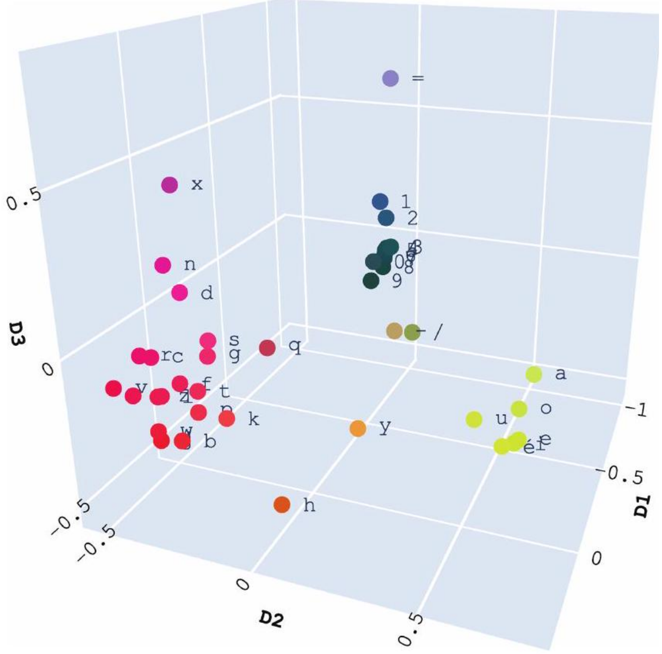
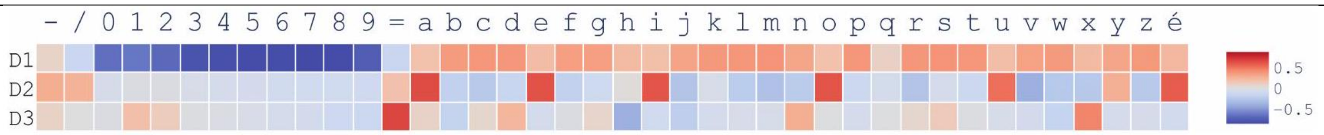
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

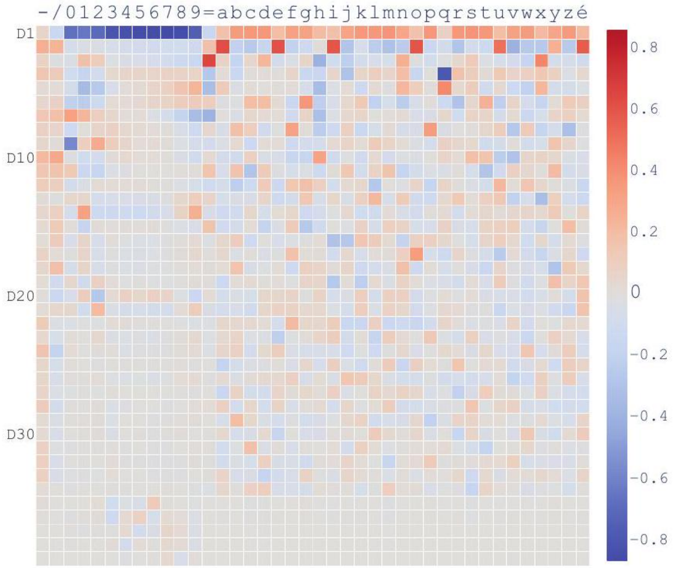
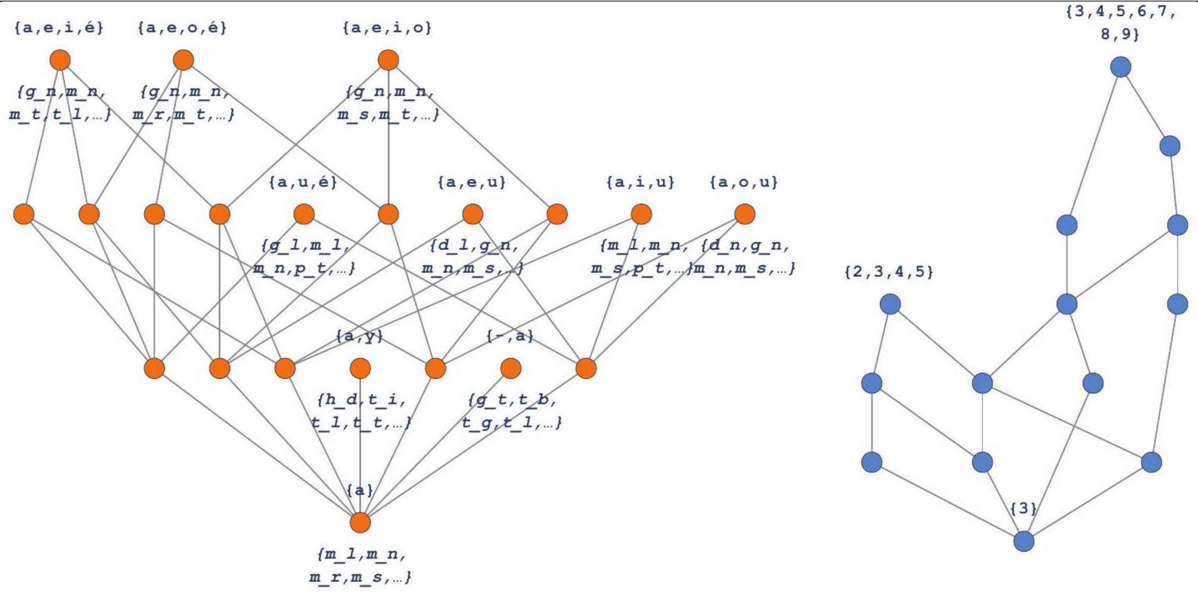
Tables

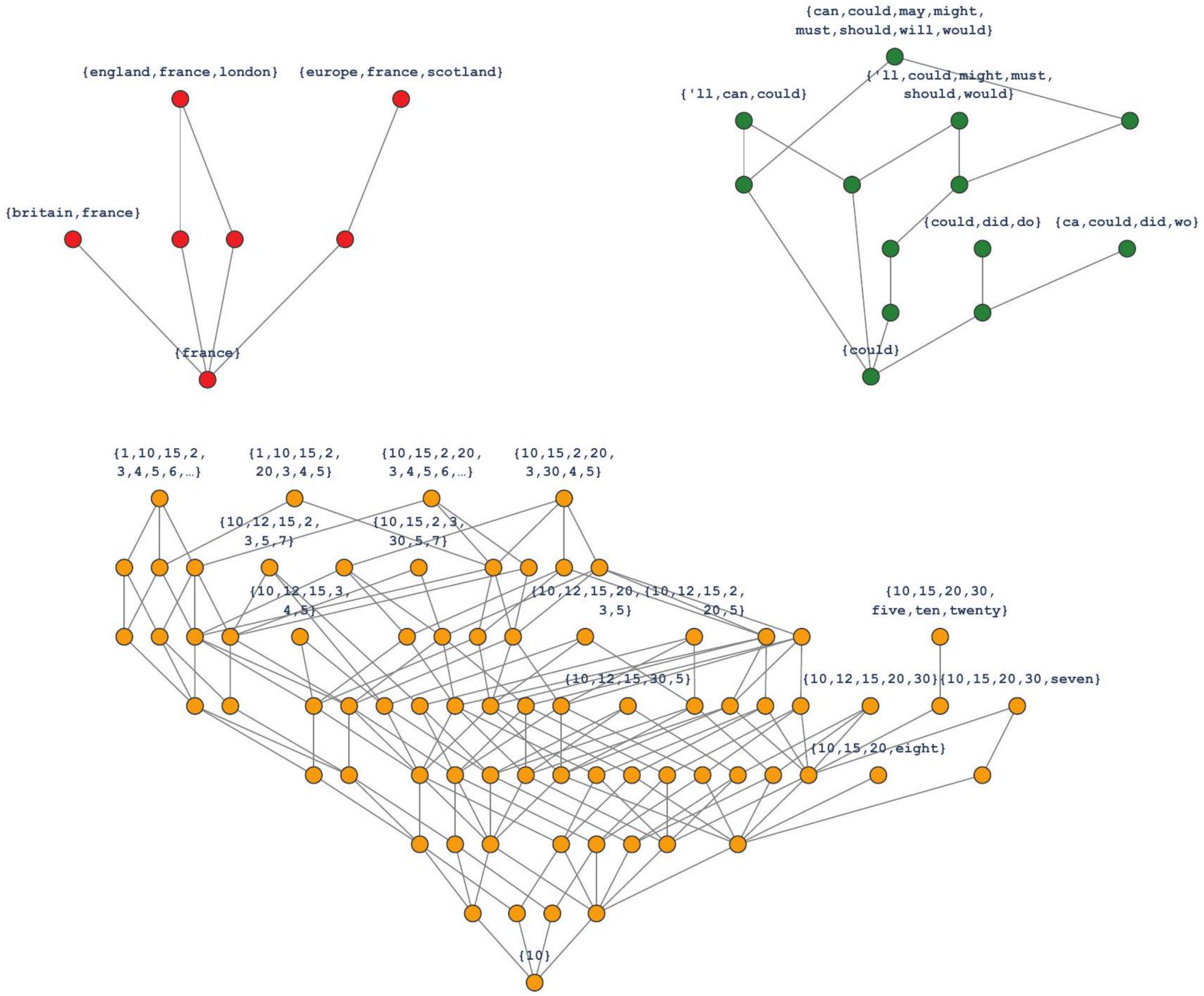
Identifier	Page	Content (LaTeX source if any)	Scan image
BradleyGastaldiTerilla2023_TAB_001	009	(no LaTeX source; 1722×594 px region — see tables.html)	—

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with `pdfdrill vision` (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
BradleyGastaldiTerilla2023_DIA_0001	001		tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0002	005	$\begin{tikzcd} & k^Y \\ X \arrow[ur] \arrow[r, hook] & k^X \arrow[u, dashed, "M^*"'] \end{tikzcd}$	tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0003	005	$\begin{tikzcd} k^Y \arrow[d, dashed, "M"'] & Y \arrow[l, hook'] \arrow[dl] \\ k^X & \end{tikzcd}$	tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0004	006	$\begin{tikzcd} & (\mathsf{Set}^{\mathsf{D}})^{op} \\ \mathsf{C} \arrow[ur] \arrow[r, hook, "Yoneda"'] & \mathsf{Set}^{\mathsf{C}^{op}} \arrow[u, dashed, "F^*"'] \end{tikzcd}$	tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0005	006	$\begin{tikzcd} (\mathsf{Set}^{\mathsf{D}})^{op} \arrow[d, dashed, "F_*"'] & \mathsf{D} \arrow[l, hook', "Yoneda"'] \arrow[dl] \\ \mathsf{Set}^{\mathsf{C}^{op}} & \end{tikzcd}$	tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0006	007		tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0007	008		tex.zip (local)

Identifier	Page	Image	Source
BradleyGastaldiTerilla2023_DIA_0008	008		tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0009	009		tex.zip (local)

Identifier	Page	Image	Source
BradleyGastaldiTerilla2023_DIA_0010	010		tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0011	012		tex.zip (local)
BradleyGastaldiTerilla2023_DIA_0012	012		tex.zip (local)

Identifier	Page	Image	Source
BradleyGastaldiTerilla2023_[illegible]_0013	012		tex.zip (local)